SOFTWARE ENGINEERING

LAB 7 – FUZZ TESTING

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CLASS: 5K

SS1:

```
PS C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7> python .\income_tax_parser.py
==== Running clean_pan ====
Enter a PAN (e.g. abcde 1234f): abcde1234h
Cleaned PAN: ABCDE1234H

==== Running parse_amount ====
Enter an amount (e.g. Rs 12,50,000): 2500000
Amount in rupees: 2500000

==== Running parse_tax_slab ====
Enter a tax slab as LOW-HIGH:RATE (e.g. 250000-500000:5): 2000000-3000000:10
Tax slab (low, high, rate): (2000000, 3000000, 10)
```

SS2:

```
PS C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7> python -m pytest .\test_income_tax_parser.py
============================= test session starts =============================
platform win32 -- Python 3.15.0, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7
plugins: hypothesis-6.170.15
collected 3 items

test_income_tax_parser.py FFF                                            [100%]

================================== FAILURES ===================================
_____________________________ test_clean_pan_no_crash _____________________________

s = None

    @given(st.text() | st.none())
    def test_clean_pan_no_crash(s):
        try:
>           clean_pan(s)

test_income_tax_parser.py:7:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

pan = None

    def clean_pan(pan):
        """
        Removes spaces from a PAN and upper-cases it.
        Assumes pan is a non-empty string.
        """
>       return pan.strip().replace(' ', '').upper()
               ^^^^^^^^
E       AttributeError: 'NoneType' object has no attribute 'strip'

income_tax_parser.py:6: AttributeError

During handling of the above exception, another exception occurred:

    @given(st.text() | st.none())
>   def test_clean_pan_no_crash(s):
                    ^^^
```

```
test_income_tax_parser.py:5:
_ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _ _

s = None

    @given(st.text() | st.none())
    def test_clean_pan_no_crash(s):
        try:
            clean_pan(s)
        except Exception as e:
>           assert False, f"clean_pan crashed with: {e!r} on input {s!r}"
E           AssertionError: clean_pan crashed with: AttributeError("'NoneType' object has no attribute 'strip'") on input None
E           assert False
E           Failing test case: test_clean_pan_no_crash(
E               s=None,
E           )
E           Explanation:
E               These lines were always and only run by failing test cases:
E                   C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py:8

test_income_tax_parser.py:9: AssertionError
_____________________________________________ test_parse_amount_no_crash ______________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py", line 12, in test_parse_amount_no_crash
  |     def test_parse_amount_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py", line 14, in test_parse_amount_no_crash
    |     parse_amount(s)
    |     ^^^^^^^^^^^^^^^
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\income_tax_parser.py", line 14, in parse_amount
    |     return int(text)
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py", line 16, in test_parse_amount_no_crash
    |     assert False, f"parse_amount crashed with: {e!r} on input {s!r}"
    | AssertionError: parse_amount crashed with: ValueError("invalid literal for int() with base 10: ''") on input ''
    | assert False
    | Failing test case: test_parse_amount_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py:15
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py", line 14, in test_parse_amount_no_crash
    |     parse_amount(s)
    |     ^^^^^^^^^^^^^^^
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\income_tax_parser.py", line 13, in parse_amount
    |     text = text.replace('Rs', '').replace(',', '').strip()
    |            ^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'replace'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py", line 16, in test_parse_amount_no_crash
    |     assert False, f"parse_amount crashed with: {e!r} on input {s!r}"
    | AssertionError: parse_amount crashed with: AttributeError("'NoneType' object has no attribute 'replace'") on input None
    | assert False
    | Failing test case: test_parse_amount_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py:15
    +------------------------------------
_____________________________________________ test_parse_tax_slab_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py", line 19, in test_parse_tax_slab_no_crash
  |     def test_parse_tax_slab_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py", line 21, in test_parse_tax_slab_no_crash
    |     parse_tax_slab(s)
```

```
    |     parse_tax_slab(s)
    |     ~~~~~~~~~~~~~~~^^^
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\income_tax_parser.py", line 22, in parse_tax_slab
    |     low, high = limits.split('-')
    |     ^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py", line 23, in test_parse_tax_slab_no_crash
    |     assert False, f"parse_tax_slab crashed with: {e!r} on input {s!r}"
    | AssertionError: parse_tax_slab crashed with: ValueError('not enough values to unpack (expected 2, got 1)') on input ':'
    | assert False
    | Failing test case: test_parse_tax_slab_no_crash(
    |     s=':',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py", line 21, in test_parse_tax_slab_no_crash
    |     parse_tax_slab(s)
    |     ~~~~~~~~~~~~~~~^^^
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\income_tax_parser.py", line 21, in parse_tax_slab
    |     limits, rate = line.split(':')
    |     ^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py", line 23, in test_parse_tax_slab_no_crash
    |     assert False, f"parse_tax_slab crashed with: {e!r} on input {s!r}"
    | AssertionError: parse_tax_slab crashed with: ValueError('not enough values to unpack (expected 2, got 1)') on input ''
    | assert False
    | Failing test case: test_parse_tax_slab_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py", line 21, in test_parse_tax_slab_no_crash
    |     parse_tax_slab(s)
    |     ~~~~~~~~~~~~~~~^^^
```

```
PS C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7> python -m pytest .\test_income_tax_parser.py
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py", line 23, in test_parse_tax_slab_no_crash
    |     assert False, f"parse_tax_slab crashed with: {e!r} on input {s!r}"
    | AssertionError: parse_tax_slab crashed with: ValueError('not enough values to unpack (expected 2, got 1)') on input ''
    | assert False
    | Failing test case: test_parse_tax_slab_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py", line 21, in test_parse_tax_slab_no_crash
    |     parse_tax_slab(s)
    |     ~~~~~~~~~~~~~~~^^^
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\income_tax_parser.py", line 21, in parse_tax_slab
    |     limits, rate = line.split(':')
    |                    ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7\test_income_tax_parser.py", line 23, in test_parse_tax_slab_no_crash
    |     assert False, f"parse_tax_slab crashed with: {e!r} on input {s!r}"
    | AssertionError: parse_tax_slab crashed with: AttributeError("'NoneType' object has no attribute 'split'") on input None
    | assert False
    | Failing test case: test_parse_tax_slab_no_crash(
    |     s=None,  # or any other generated value
    | )
    +-----------------------------------
=========================================================== short test summary info ============================================================
FAILED test_income_tax_parser.py::test_clean_pan_no_crash - AssertionError: clean_pan crashed with: AttributeError("'NoneType' object has no attribute 'strip'"
) on input None
FAILED test_income_tax_parser.py::test_parse_amount_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_income_tax_parser.py::test_parse_tax_slab_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
============================================================ 3 failed in 3.87s =============================================================
PS C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7>
```

SS3:

```
PS C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7> python .\income_tax_parser.py
==== Running clean_pan ====
Enter a PAN (e.g. abcde 1234f): abcde 1234h
Cleaned PAN: ABCDE1234H

==== Running parse_amount ====
Enter an amount (e.g. Rs 12,50,000): 10,00,000
Amount in rupees: 1000000

==== Running parse_tax_slab ====
Enter a tax slab as LOW-HIGH:RATE (e.g. 250000-500000:5): 1000000-1500000:5
Tax slab (low, high, rate): (1000000, 1500000, 5)
```

SS4:

```
PS C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7> python -m pytest .\test_income_tax_parser.py
============================= test session starts ==============================
platform win32 -- Python 3.15.0, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\College\sem-v\se\lab6\L6_L7_57_income_tax\LAB7
plugins: hypothesis-6.170.15
collected 3 items

test_income_tax_parser.py ...                                            [100%]

============================== 3 passed in 2.52s ===============================
```